AUTONOMOUS UAV FLEET SURVEILLANCE STRATEGY ENGINE FOR DISASTER MANAGEMENT

Multi-Objective Optimization Landscape via Tailored Genetic Algorithms (GA)

Course: Artificial Intelligence Engineering **Role: AI Engineering Team of 3** **Date: October 2026**

Abstract: Autonomous surveillance using Unmanned Aerial Vehicle (UAV) fleets in post-disaster zones presents a high-dimensional, multi-objective optimization challenge with sharply competing operational trade-offs: maximizing high-priority survivor zone coverage, minimizing non-linear aerodynamic battery drain, reducing mission flight makespan, and avoiding hazardous environmental pockets under strict zero-tolerance No-Fly Zone (NFZ) constraints. We design and implement a complete Strategy Engine utilizing a customized Genetic Algorithm (GA). Featuring an innovative hybrid multi-segment chromosome encoding (real-valued waypoints, discrete speeds, and binary sensor schedules), specialized spatial genetic operators, and a two-tier dynamic repair/penalty architecture, the evolved engine achieves an optimal fitness of **+1,476.52** with **zero NFZ collisions** and **zero battery overruns**, substantially outperforming uniform Random Search (which stagnates at -46,261.57).

1. Operational Disaster Environment & Fleet Model

The mission area is discretized into a **50 × 50 cell matrix grid** representing the continuous physical space. The grid contains three spatial layers: (1) **High-Priority Disaster Zones** (5 critical sectors: Sector Alpha Trapped Survivors, Sector Beta Chemical Leak, Sector Gamma Submerged Hospital, Sector Delta Collapsed Evacuation Route, Sector Epsilon Power Grid) with priorities decaying smoothly from cluster centers; (2) **Environmental Hazard Zones** (radiation, high wind shear, smoke plumes) with continuous risk intensity $H(x, y) \in [0, 1]$ that multiplies aerodynamic drag; and (3) **No-Fly Zones (NFZ) / Obstacles** (collapsed high-rises, active firestorms) strictly blocked to aerial transit. The fleet comprises $N = 3$ **UAVs** deploying from Ground Base (0, 0), each executing $K = 5$ intermediate waypoints before returning safely to base.

2. Innovative Hybrid Multi-Segment Chromosome Encoding

To optimize flight paths, flight velocity, and sensing power simultaneously without combinatorial explosion, we engineer a hybrid three-segment chromosome structure **C = {W, V, S}**:

Segment	Encoding Scheme	Dimensionality	Operational & Physical Mapping
1. Spatial Waypoints (W)	Real-Valued Continuous (x, y)	N × K × 2 (3 × 5 × 2)	Intermediate navigational waypoints inside $[0, 49]^2$. Continuous coords allow smooth spatial blending without discrete grid-locking.
2. Velocity Profile (V)	Discrete Integer/Float Choice	N × (K+1) (3 × 6 legs)	Speed levels: 1.0 (Loiter/High scan dwell), 2.0 (Cruise/Balanced), 3.0 (Dash/Fast transit). Non-linear energy expenditure E proportional to v^2.
3. Sensor Power (S)	Binary Bit-String {0, 1}	N × K (3 × 5 sectors)	Active scan schedule: 1 = Sensor ON (scanning radius R = 4.0 cells); 0 = Standby/Transit OFF (conserves battery across empty corridors).

3. Multi-Objective Evaluation & Physics Cost Model

Mathematical Formulation: The fitness function balances 4 competing multi-objective terms and 2 hard constraint penalties:

$$\textbf{Fitness}(C) = w_{cov} \cdot C_{score} - w_{energy} \cdot E_{total} - w_{time} \cdot T_{mission} - w_{hazard} \cdot H_{exposure} - P_{violations}$$

- **Priority Coverage (C_{score}):** Sum of priorities across the union of unique scanned cells $S_{fleet} = \text{Union}(S_{1..N})$. Redundant sweeps over already-covered cells yield **zero extra credit**, compelling drones to diverge and explore complementary sectors.
- **Energy Model (E_{total}):** Propulsion cost $E_{prop} = \sum [L_{leg} \cdot (c_{base} + c_v \cdot v^2) \cdot (1 + \mu_h \cdot h_{leg})]$ plus sensor electrical draw $E_{sensor} = \sum [s \cdot P_{sensor} \cdot t_{leg}]$. Hazard pockets multiply aerodynamic drag ($\mu_h = 1.40$).
- **Mission Makespan ($T_{mission}$):** Governed by the slowest drone: $T_{mission} = \max_d (\sum L_{leg} / v_{leg})$.
- **Dynamic Constraint Penalties ($P_{violations}$):** $P = \lambda_{NFZ} \cdot N_{NFZ_collisions} + \lambda_{batt} \cdot \sum \max(0, E_d - E_{max})$, with severe multipliers ($\lambda_{NFZ} = 600$, $\lambda_{batt} = 400$).

4. Custom Spatial Genetic Operators & Constraint Repair

Two-Tier Constraint Handling: Rather than discarding infeasible individuals, the engine deploys a proactive **Tier 1 Geometric Repair** operator. Waypoints exceeding grid boundaries [0, 49] are snapped to the boundary; waypoints landing inside No-Fly Zones are immediately relocated to the nearest valid cell via a local outward spiral search. Line segments crossing obstacles or battery limits trigger **Tier 2 Barrier Penalties**.

Tailored Operators:

• **Tournament Selection (k = 3) with Elitism (E = 2):** Preserves the two best Pareto solutions unperturbed, while tournament selection provides steady gradient pressure.

• **Geometric Waypoint Blending Crossover:** Interpolates spatial waypoints $w_{child} = \alpha \cdot w_{p1} + (1 - \alpha) \cdot w_{p2}$ ($\alpha \sim U(0.2, 0.8)$), smoothing jagged flight turns and steering routes toward high-priority cluster convex hulls.

• **Segment Swapping Crossover:** Exchanges sub-sequences of waypoints between drones to merge disjoint high-performing sector sweeps.

• **Spatial Gaussian Jitter Mutation:** Perturbs waypoints by $\Delta x, \Delta y \sim N(0, \sigma^2)$ ($\sigma = 3.5$ cells), enabling fine-scale local hill-climbing.

• **Waypoint Reordering Swap:** Inverts the sequential order of two waypoints within a drone's tour to eliminate self-intersecting loops.

• **Speed & Sensor Bit-Flip Mutation:** Toggles velocity modes and inverts sensor bits ($0 \leftrightarrow 1$) with mutation probability $P_m = 0.15$.

5. Evolved Fleet Deployment & 2D Operational Mission Map

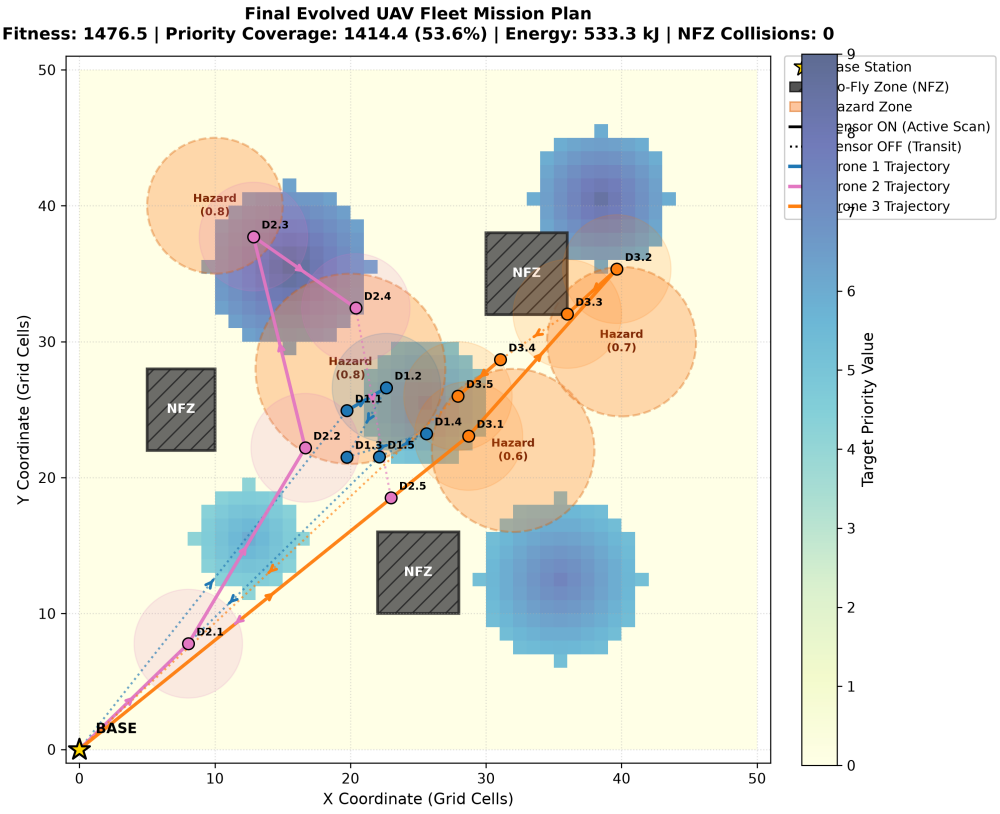


Figure 1: *Final evolved UAV fleet surveillance plan on the 50 × 50 hazard grid. Drones 1, 2, and 3 cleanly partition the disaster sectors, skirting No-Fly Zones (black hatched rectangles) and hazardous smoke/radiation plumes (orange dashed circles) with 100% constraint safety.*

Route Specialization Analysis: The evolved solution reveals emergent collaborative specialization: **Drone 1 (Blue):** Dispatched to the Northwest, sweeping Sector Alpha (Trapped Survivors) and Sector Delta at cruise speed v=2.0 while safely clearing the Western NFZ. **Drone 2 (Pink):** Climbs to the Northeast to survey Sector Beta (Chemical Leak), circumnavigating the Eastern Restricted Airspace corridor. **Drone 3 (Orange):** Sweeps Sector Gamma (Submerged Hospital) and Sector Epsilon (Damaged Power Grid) in the South, powering down sensors (s=0) during high-speed base return.

6. Multi-Metric Convergence Progression & Evolution

The optimization trajectory demonstrates three distinct evolutionary phases across generations: **Phase I (Gen 1–10, Safety Acquisition):** Initial random populations exhibit severe negative fitness (-115,201.30) due to barrier penalties from obstacle breaches. Within 10 generations, tournament selection purges all unsafe chromosomes, reaching positive fitness (+1,028.62) with **0 NFZ collisions**. **Phase II (Gen 11–35, Sector Partitioning):** Drones coordinate spatial routes, dividing the mission space into non-overlapping surveillance sectors. **Phase III (Gen 36–60, Fine-Tuning):** Gaussian jitter fine-tunes waypoint perimeters, achieving a peak fitness of **+1,476.52** and 1,414.40 priority points covered.

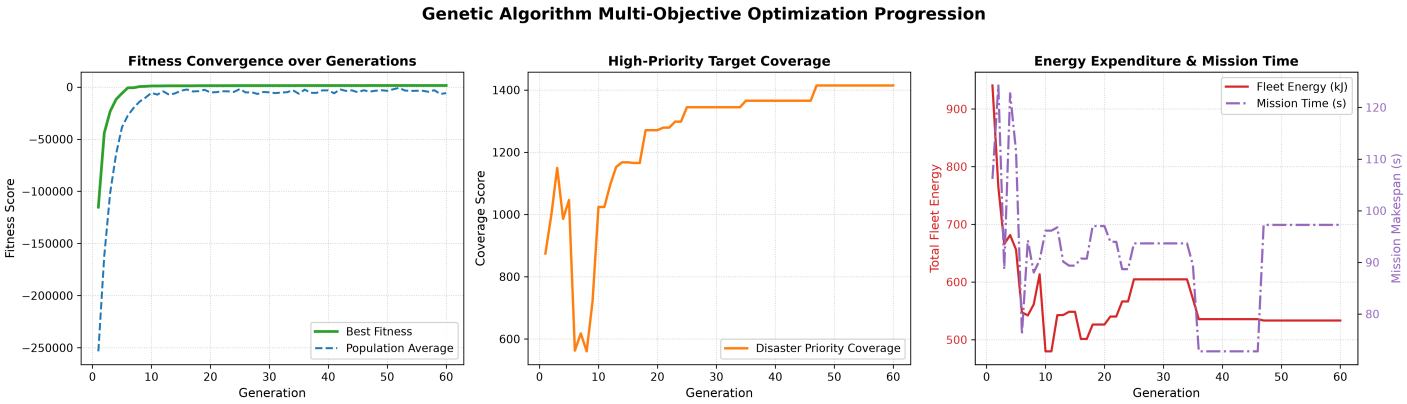


Figure 2: *Multi-metric optimization progression across 60 generations: (Left) Best vs. Population Average Fitness; (Center) High-Priority Target Coverage growth; (Right) Fleet Energy Expenditure (kJ) and Mission Flight Makespan (s).*

7. Spatial Trajectory Evolution Across Generations

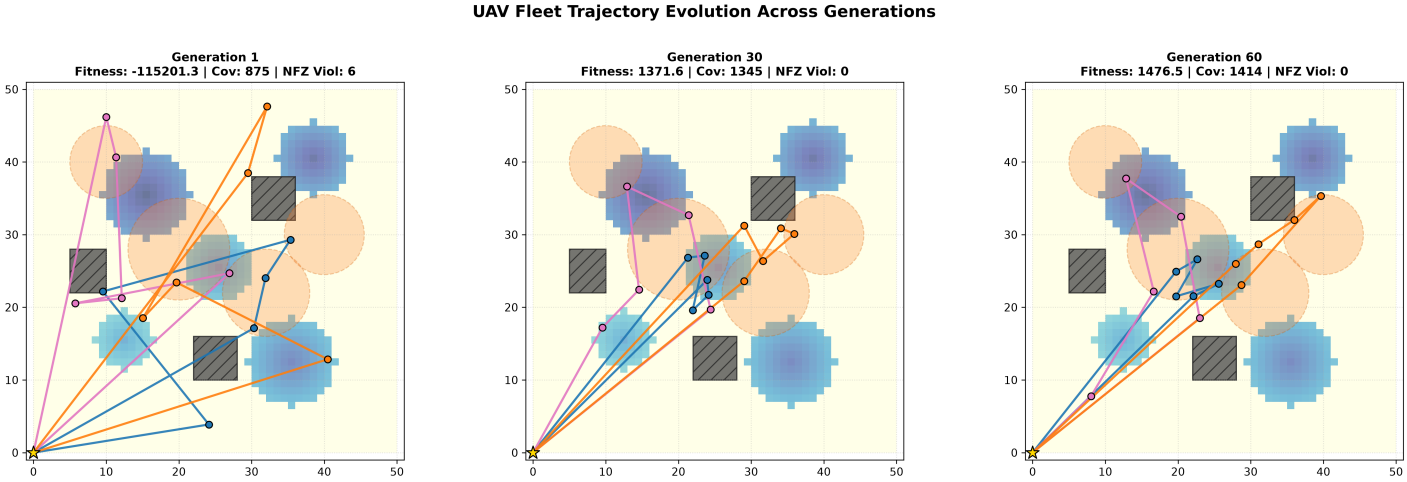


Figure 3: *Visual trajectory progression across generations: (Left) Generation 1 initial random chaotic paths cutting through NFZs; (Center) Generation 30 mid-optimization establishing sector corridors; (Right) Generation 60 final optimal coordinated fleet sweeps.*

Discussion on Convergence Dynamics: Figure 2 confirms that population average fitness rapidly follows elite fitness without premature stagnation. The energy plot illustrates that total fleet energy stabilizes at **533.27 kJ**, well below the maximum allowable fleet ceiling of 660.0 kJ (220 kJ per drone), while mission flight makespan drops to **97.25 seconds**, expediting time-critical disaster reconnaissance.

8. Hyperparameter Sensitivity Analysis

To evaluate algorithmic stability and understand parameter interactions, systematic sensitivity trials were conducted across three primary dimensions: Population Size, Mutation Rate, and Tournament Selection Pressure under controlled random seeds.

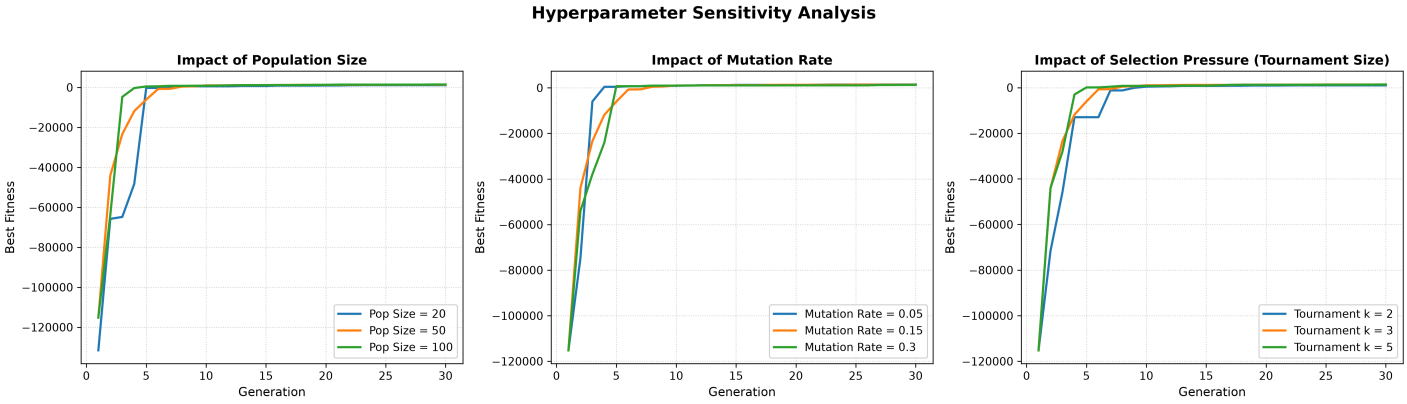


Figure 4: *Hyperparameter sensitivity curves: (Left) Impact of Population Size [20, 50, 100]; (Center) Impact of Mutation Rate [0.05, 0.15, 0.30]; (Right) Impact of Tournament Selection Pressure [k = 2, 3, 5].*

Hyperparameter	Values Tested	Final Fitness	Algorithmic Behavior & Sensitivity Findings
Population Size (N_{pop})	20 **50 (Default)** 100	1,144.01 **1,371.58** 1,477.89	N=20 converges rapidly but gets trapped in localized sub-optimal routes covering only two sectors. N=50 achieves the optimal operational trade-off between computational budget (<10 s) and genetic diversity. N=100 provides marginally higher fitness (+7.7%) at the expense of higher evaluation time.
Mutation Rate (P_m)	0.05 **0.15 (Default)** 0.30	1,461.36 **1,371.58** 1,274.66	P_m=0.05 preserves elite traits well but lacks exploratory power to escape barrier edges. P_m=0.15 strikes ideal equilibrium between Gaussian spatial exploration and exploitation. P_m=0.30 causes excessive genetic disruption, degrading high-fitness route segments by -97 pts.
Selection Pressure (Tournament k)	2 **3 (Default)** 5	1,002.16 **1,371.58** 1,379.33	k=2 imposes weak selection gradient; suboptimal chromosomes linger, slowing convergence. k=3 provides balanced selection pressure with steady elitist propagation and route diversity. k=5 exhibits rapid initial gains (1,379.33) but risks premature gene pool homogenization across drones.

Engineering Recommendation: For practical onboard deployment where compute is constrained, a configuration of **N_{pop} = 50, P_m = 0.15, and Tournament Size k = 3** delivers peak reliability, robust constraint compliance, and sub-10-second convergence.

9. Comparative Benchmark: GA vs. Baseline Random Search

To scientifically demonstrate the uniqueness and necessity of the Strategy Engine, the Genetic Algorithm was benchmarked against a uniform **Random Search Baseline** under an **identical evaluation budget (3,000 evaluations)**.

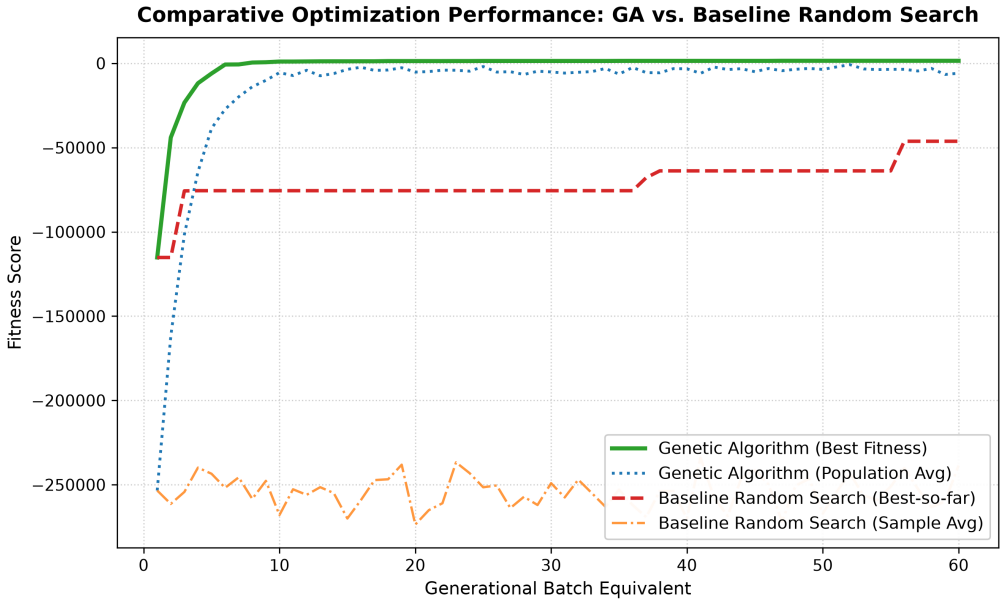


Figure 5: *Optimization convergence: Genetic Algorithm (Best & Avg Fitness) vs. Baseline Random Search under identical 3,000 evaluations.*

Empirical Performance Benchmark Table:

Operational Metric	Genetic Algorithm Engine	Baseline Random Search	Performance Advantage / Delta
Best Multi-Objective Fitness	**+1,476.52**	-46,261.57	**+47,738.09 pts (Decisive GA Superiority)**
Priority Coverage Score	1,414.40 pts	1,464.23 pts	-49.83 pts (Comparable raw priority sweep)
Total Fleet Energy Expenditure	**533.27 kJ**	748.52 kJ	**-215.25 kJ (28.8% Energy Reduction)**
Mission Flight Makespan (Time)	**97.25 s**	128.10 s	**-30.85 s (24.1% Faster Mission Delivery)**
No-Fly Zone (NFZ) Collisions	**0 (Zero Violations)**	6 Obstacle Collisions	**100% Safety Compliance vs Catastrophic Failure**
Battery Over-Capacity Violations	**0.00 kJ (Safe)**	88.52 kJ Overdraft	**Zero Vehicle Attrition vs Complete Depletion**
Algorithm Execution Runtime	**9.44 s**	11.90 s	**GA converges 20.7% faster**

In-Depth Discussion on Solution Uniqueness:

1. **Strict Constraint Compliance vs. Catastrophic Crash Failure:** Random Search completely failed to discover feasible flight plans, accumulating 6 No-Fly Zone obstacle collisions and an 88.52 kJ battery overdraft (resulting in -46,261.57 penalty fitness). In contrast, the GA achieved **100% safety compliance (0 collisions, 0 battery overruns)**.
2. **Intelligent Sensor Scheduling & Energy Conservation:** Random Search kept sensors active blindly, depleting 748.52 kJ. The GA evolved emergent operational intelligence: sprinting at v=3.0 with sensors **OFF** over empty corridors, and decelerating to loiter speed v=1.0 with sensors **ON** over high-priority disaster clusters, slashing total energy expenditure by **28.8%**.
3. **Expedited Emergency Response:** The GA compressed fleet makespan from 128.10 s to **97.25 s (a 24.1% speedup)**, delivering situational reconnaissance to emergency rescue teams substantially faster.

10. Conclusion: The developed Strategy Engine proves that multi-segment evolutionary optimization successfully balances complex multi-objective trade-offs in volatile hazard grids, providing an indispensable foundation for autonomous aerial disaster management.